Does Air Pollution Shrink the Shopping Map?

Card spending in Seoul by distance from home — batches 1–2

APC project team

Working draft, 22 October 2026

Question and answer

Question. On polluted days, do Seoul residents spend less, and closer to home?

Answer so far.

- Pollution is one city-wide daily series (96% of the variance) $\Rightarrow$ start with the city as a time series.
- City OLS (levels, Newey–West): -0.8% spending per 10 $\mu g/m^3$, mostly online; shares and distance flat.
- Instruments disagree: wind and upwind$\times$wind are weak in the city series; inversions ($F = 8$) say online *falls*; panel wind IV says online *rises*; all move spending on days *before* the shock.
- Within a residence-day, offline spending **never** moves toward home.
- Robust: **no offline contraction**, small level effects; online response unresolved.

City as one time series

City series: OLS in levels (main)

$$Y_t = \beta\,\text{PM2.5}_t + X_t'\gamma + \varepsilon_t \qquad (t = \text{day};\ \text{Newey–West, 7 lags})$$

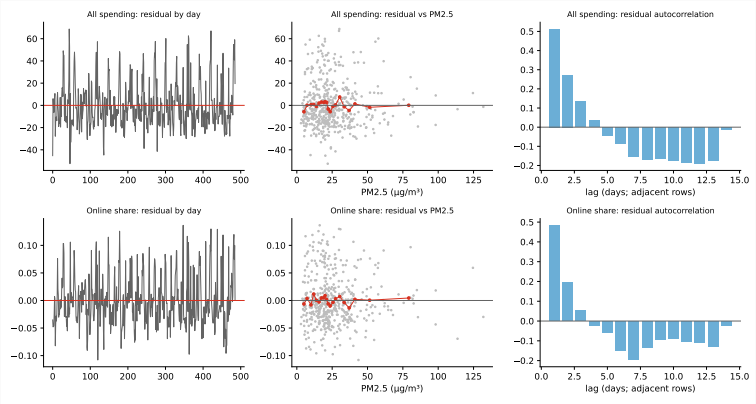
$$\text{PPML: } E[Y_t] = \exp(\beta \log \text{PM2.5}_t + X_t'\gamma)$$

Outcome	Mean	OLS per 10 μg/m^3	% of mean	PPML elasticity
All spending (bn won)	206.1	-1.63** (0.75)	-0.8%	-0.012
Offline (bn won)	175.2	-0.59 (0.36)	-0.3%	-0.002
Online (bn won)	30.9	-1.04* (0.60)	-3.4%	-0.084
Own dong (bn won)	46.8	-0.07 (0.14)	-0.1%	0.003
Ring 3+ (bn won)	70.9	-0.33* (0.17)	-0.5%	-0.006
Online share (pp)	14.2	-0.31 (0.22)		
Home share (pp)	37.2	-0.19 (0.13)		
Mean km, offline (km)	4.7	-0.005 (0.007)		

486 days. Y_t: spending of Seoul residents (bn won) or share; PM2.5$_t$: city mean (μg/m^3; 10 $\approx$ 0.6 SD). X_t: day of week, year-month, holidays, cubic in temperature, max/min temperature, rain, precipitation, humidity, wind speed. Logs/PPML elasticities ≈ 0.

Residuals

$$\hat{\varepsilon}_t = Y_t - \hat{\beta}\,\text{PM2.5}_t - X_t'\hat{\gamma}$$



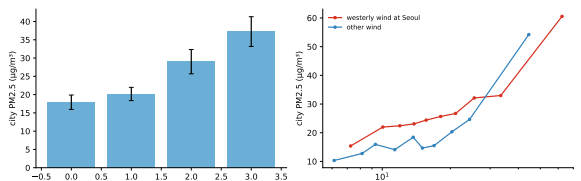
Flat in PM2.5; autocorrelation 0.5 at lag 1, gone by day 4 $\Rightarrow$ Newey–West (7 lags).

Three instruments for the city series

$$Y_t = \beta\,\mathrm{PM2.5}_t + X_t'\gamma + \varepsilon_t, \qquad \mathrm{PM2.5}_t = Z_t'\pi + X_t'\delta + \nu_t \quad \text{(2SLS)}$$

Z_t: wind sectors; westerly dummy; inversions (count at 21/03/09 KST, 03 KST strength); $\log \mathrm{PM2.5}_{t-1}^{\text{Baengnyeong}} \times \mathrm{west}_t$ (with main effects in X_t)

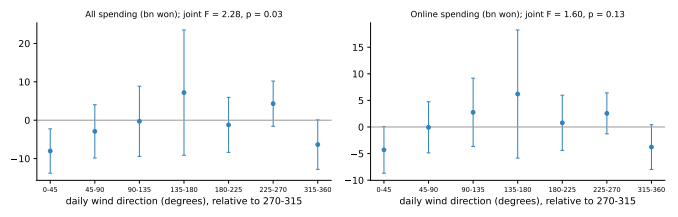
	All	Offline	Online	Online share	Home share	Mean km	FS F	J p
OLS	-1.63**	-0.59	-1.04*	-0.308	-0.188	-0.005		
	(0.75)	(0.36)	(0.60)	(0.217)	(0.128)	(0.007)		
Wind sectors (8, receptor-only)	-3.02	1.98	-5.01	-1.949	-1.233	-0.052	1.5	0.04
	(6.95)	(3.46)	(5.07)	(1.678)	(1.128)	(0.045)		
Westerly wind	16.59	13.50	3.10	0.344	0.128	0.048	2.4	
	(16.57)	(10.80)	(8.34)	(2.645)	(1.729)	(0.080)		
Thermal inversions	-9.22**	-1.93	-7.29**	-2.215**	-1.555**	-0.022	8.0	0.28
	(4.23)	(1.90)	(3.04)	(0.961)	(0.653)	(0.026)		
Upwind PM2.5 × westerly	-3.26	-1.14	-2.12	-1.221	-1.149	0.045	2.6	
	(8.22)	(3.83)	(5.60)	(1.993)	(1.405)	(0.062)		
Inversions + upwind	-9.06	-1.05	-8.01	-2.775	-2.266*	0.029	2.2	0.47
	(7.37)	(3.00)	(5.54)	(1.853)	(1.319)	(0.048)		



Reduced form: city spending on the instruments

$$Y_t = Z_t'\theta + X_t'\gamma + \varepsilon_t \qquad \text{(OLS, 486 days, Newey–West 7 lags)}$$

	Wind sectors joint F [p]	Westerly dummy	Inversions count	Inversions strength (°C)	Upwind PM2.5 × westerly
All spending (bn won)	2.28 [0.025]	3.53 (2.51)	-2.06** (1.01)	-1.47 (1.04)	-1.49 (3.58)
Offline (bn won)	2.40 [0.019]	2.87*** (1.10)	-1.12** (0.51)	0.02 (0.50)	-0.52 (1.68)
Online (bn won)	1.60 [0.131]	0.66 (1.71)	-0.93 (0.78)	-1.50** (0.68)	-0.97 (2.48)
Online share (pp)	1.29 [0.250]	0.073 (0.562)	-0.267 (0.265)	-0.463** (0.216)	-0.558 (0.835)
Home share (pp)	1.22 [0.289]	0.027 (0.368)	-0.150 (0.172)	-0.343** (0.144)	-0.525 (0.545)
Mean km, offline	1.97 [0.055]	0.010 (0.016)	-0.005 (0.008)	-0.004 (0.007)	0.021 (0.026)



Wind direction moves spending (joint $p = 0.03$ all, 0.02 offline) although it barely moves city PM2.5 (first-stage $F = 1.5$): consistent with direct effects of wind on spending.

Checks from the new papers

Weather proxies (Carson & Yu 2026): city 2SLS above with X_t = city-mean weather, + central station, station only, or tails trimmed — stable.

	OLS		2SLS, inversions		
	All spending	Online share	All spending	Online	Online share
City-mean weather (main)	-1.63** (0.75)	-0.308 (0.217)	-9.22** (4.23)	-7.29** (3.04)	-2.215** (0.961)
City mean + central station (two proxies)	-1.89*** (0.72)	-0.350 (0.213)	-8.71*** (3.36)	-7.02*** (2.45)	-2.185*** (0.773)
Central station only	-1.65** (0.73)	-0.272 (0.216)	-8.88** (3.69)	-6.91*** (2.67)	-2.059** (0.840)
Main, excluding the 5% coldest and warmest days	-1.73** (0.76)	-0.319 (0.224)	-9.35** (4.69)	-7.64** (3.44)	-2.319** (1.080)

Conley (1999), panel $S_{rt} = \beta \log \text{PM2.5}_{rt} + \alpha_r + X'_{rt}\gamma + \varepsilon_{rt}$: SE rises with the cutoff; at city scale = date clustering.

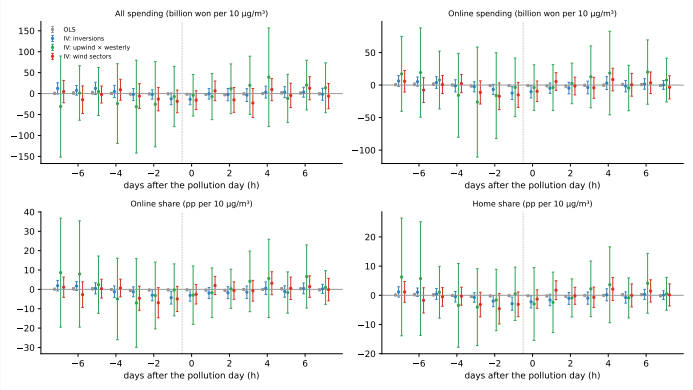
	Home share		Online share		Mean km, offline	
	OLS	2SLS	OLS	2SLS	OLS	2SLS
Coefficient	-0.0009	0.0142	-0.0019	0.0230	-0.0136	0.0645
Residence + date clusters (main)	0.0019 [0.635]	0.0056 [0.011]	0.0031 [0.527]	0.0086 [0.008]	0.0112 [0.225]	0.0317 [0.042]
Date clusters		0.0056 [0.011]				0.0321 [0.045]
Conley, 5 km × 7 days	0.0008 [0.246]	0.0025 [0.000]	0.0012 [0.106]	0.0036 [0.000]	0.0053 [0.011]	0.0163 [0.000]
Conley, 10 km × 7 days	0.0011 [0.407]	0.0035 [0.000]	0.0017 [0.263]	0.0052 [0.000]	0.0070 [0.052]	0.0206 [0.002]
Conley, 20 km × 7 days	0.0016 [0.555]	0.0047 [0.003]	0.0025 [0.433]	0.0073 [0.002]	0.0091 [0.136]	0.0264 [0.015]
Conley, 50 km × 7 days	0.0020 [0.643]	0.0057 [0.013]	0.0032 [0.546]	0.0091 [0.011]	0.0115 [0.239]	0.0328 [0.049]

PPML (Santos Silva & Tenreyro 2006), $E[Y_t] = \exp(\beta \log \text{PM2.5}_t + X'_t\gamma)$: online elasticity -0.084 vs. -0.045 by log-OLS; $V \propto E$ rejected (efficiency only); RESET flags all/online $\Rightarrow$ OLS levels stays main.

City local projections: leads move too

$$Y_{t+h} = \beta_h \,\mathrm{PM2.5}_t + \rho_1 \mathrm{PM2.5}_{t-1} + \rho_2 \mathrm{PM2.5}_{t-2} + X_t'\gamma_h + \varepsilon_{t+h}, \quad h = -7, \ldots, 7$$

OLS, or 2SLS with $\mathrm{PM2.5}_t$ instrumented by Z_t; $h < 0$ are placebo leads



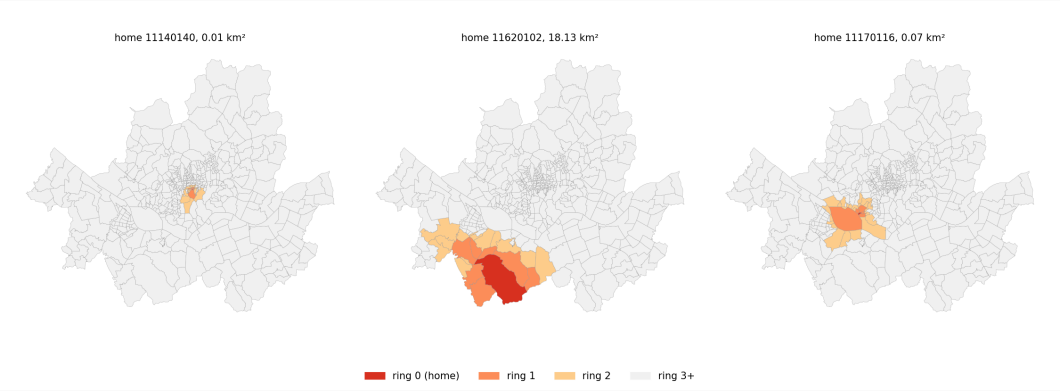
Inversion-IV online response at $t-1, t-2 \approx$ day t; vanishes without weather controls.

Neighborhood panel

Data

- **Spending**: Shinhan card, raw files, Jan–Aug 2018 and 2019 (486 days). Seoul residents of 466 legal dongs; merchants in Seoul + online. Built into a residence × merchant-dong × day flow table (150M cells from 499M raw rows).
- **Pollution**: AirKorea PM2.5, fixed set of 94 non-roadside monitors (gaps filled from neighbors, Auffhammer et al. 2013), inverse-distance average within 5 km of each dong.
- **Weather and wind**: KMA stations, same rule; daily vector-mean wind direction in 8 sectors.
- **Rings**: borders crossed from home: 0 own dong, 1 adjacent (median 6 dongs), 2 two borders (14), 3+ rest of Seoul; online separate or counted as home.

Rings around three example neighborhoods

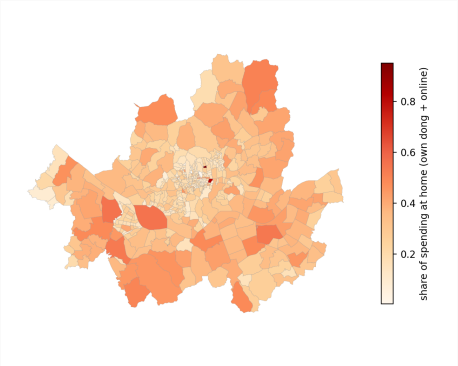


Dong areas range from 0.04 to 3.7 km^2 (p10–p90), so km bands are used as a check.

Where residents spend

	All days	Weekdays	Weekends	Share in floor cells
Own dong (ring 0)	0.227	0.219	0.250	0.025
Adjacent (ring 1)	0.182	0.172	0.209	0.120
Two borders (ring 2)	0.097	0.092	0.111	0.277
Rest of Seoul (ring 3+)	0.344	0.336	0.369	0.494
Online	0.150	0.181	0.061	0.030

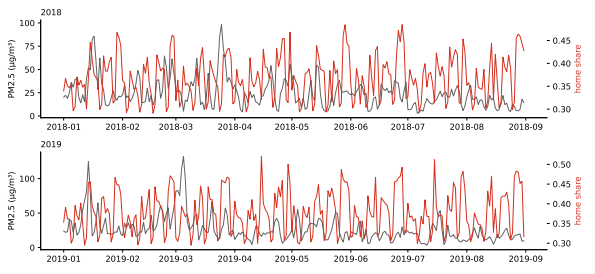
Half of ring-3+ spending sits in cells at the provider's reporting floor (5 transactions): distant flows are measured coarsely.



Pollution varies over time, not across neighborhoods

Component of the variance of log PM2.5	Share
Between days (city-wide daily mean)	0.960
Persistent across dongs (dong mean of within-day deviation)	0.013
Residual: dong-specific daily deviation	0.027
Explained by residence, year-month and day-of-week FE	0.246

Identification is a daily time series: errors are correlated across dongs within a day $\Rightarrow$ cluster by date (and by dong).



Design

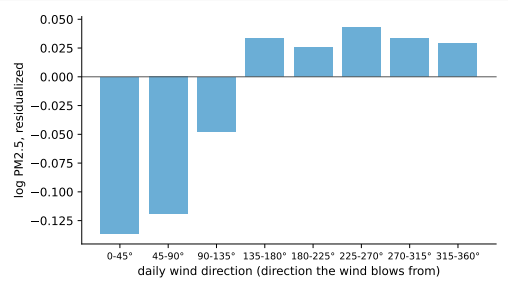
Headline: within residence r and day t, stacked over rings k (Poisson):

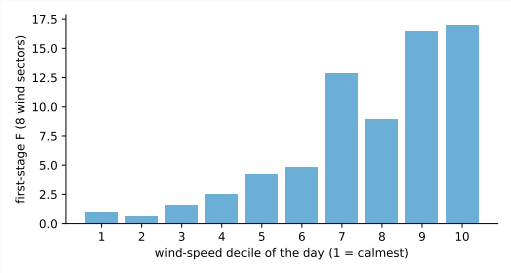
$$E[Y_{rkt}] = \exp\left(\sum_{j\geq 1}\beta_j \mathbf{1}[k=j]\,\log \text{PM2.5}_{rt} + \mu_{rt} + \psi_{rk} + X'_{rt}\lambda_k\right)$$

- μ_{rt} absorbs anything that moves a dong's total spending that day; β_j = elasticity of ring j *relative to home*.
- Controls (ring-specific): year-month, day of week, holidays, weather cells (temperature $\times$ rain $\times$ wind speed), humidity.
- **IV**: 8 wind-direction sectors (Deryugina et al. 2019); control function in Poisson, 2SLS for shares.
- **SE**: two-way clustered by dong (466) and date (486) (Cameron & Miller 2015); Anderson–Rubin sets for 2SLS.

First stage: wind moves pollution, more so on windy days

$$\log \text{PM2.5}_{rt} = \sum_{s=1}^{8} \pi_s \, \mathbf{1}[\text{WD}_{rt} = s] + \alpha_r + X'_{rt}\delta + v_{rt}$$

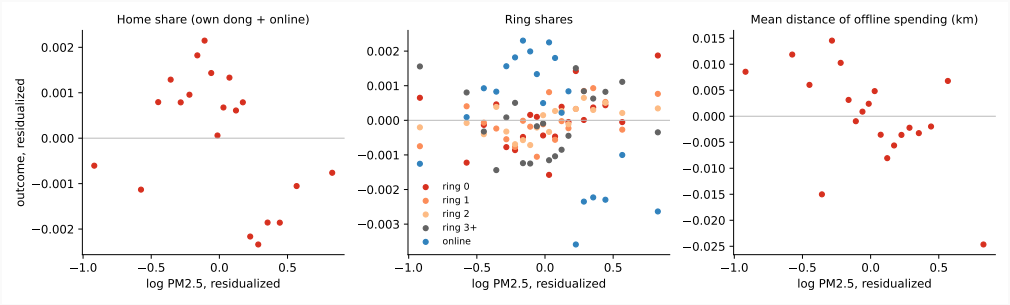




F = 6.9 (8 sectors, two-way clustered); 34 with a single westerly dummy. Stronger on windy days: transported pollution, not local conditions.

Without instruments: nothing moves

$\tilde{S}_{rt}$ vs. $\widetilde{\log \text{PM2.5}}_{rt}$, $\quad \tilde{v}$ = residual of v on $\alpha_r + X_{rt}$



Residualized on dong, year-month, day of week, holidays, weather, humidity. Slopes ≈ 0.

Headline: away-from-home spending falls relative to home + online

$$E[Y_{rkt}] = \exp\left(\sum_{j\geq 1} \beta_j \mathbf{1}[k=j] \log \text{PM2.5}_{rt} + \mu_{rt} + \psi_{rk} + X'_{rt}\lambda_k\right)$$

Y_{rkt}: spending of residents of dong r in ring k on day t; ring 0 (home) is the reference; PPML-IV adds $\mathbf{1}[k=j]\hat{v}_{rt}$ from the wind first stage

	(1) PPML	(2) PPML-IV	(3) PPML	(4) PPML-IV
Ring 1: adjacent dongs × log PM2.5	0.017	-0.124***	-0.001	0.009
	(0.012)	(0.047)	(0.002)	(0.010)
Ring 2: two borders away × log PM2.5	0.018	-0.115**	-0.001	0.020
	(0.012)	(0.045)	(0.004)	(0.013)
Ring 3+: rest of Seoul × log PM2.5	0.012	-0.092**	-0.007	0.043**
	(0.010)	(0.038)	(0.005)	(0.019)
Online × log PM2.5			-0.050	0.304**
			(0.031)	(0.119)
Online counted as	home (ring 0)		separate ring	
First-stage F (wind)		6.9		6.9
Exogeneity test p		0.027		0.044
Observations (residence-day-ring)	901,568	901,568	1,125,999	1,125,999

Cols 1–2: online counted as home. Cols 3–4: online as its own ring. Second-stage SEs (bootstrap in the final run).

Distance bands instead of rings: same picture

$$E[Y_{rbt}] = \exp\left(\sum_{j\geq 1} \beta_j \mathbf{1}[b=j] \log \text{PM2.5}_{rt} + \mu_{rt} + \psi_{rb} + X'_{rt}\lambda_b\right)$$

b: centroid distance from the home dong — own dong (reference), $\leq$2 km, 2–5 km, >5 km, online. 2 km: Seoul card data show proximity effects strongest within 2 km (Choi et al. 2024)

	(1) PPML	(2) PPML-IV	(3) PPML	(4) PPML-IV
Up to 2 km × log PM2.5	0.017	-0.142***	-0.003	-0.004
	(0.013)	(0.047)	(0.003)	(0.012)
2–5 km × log PM2.5	0.018	-0.106**	-0.001	0.028**
	(0.012)	(0.045)	(0.003)	(0.012)
More than 5 km × log PM2.5	0.011	-0.089**	-0.008	0.044**
	(0.010)	(0.037)	(0.005)	(0.020)
Online × log PM2.5			-0.049	0.304**
			(0.031)	(0.119)
Online counted as	home (own dong + online)		separate band	
Observations	895,982	895,982	1,120,413	1,120,413

As with rings: no offline contraction; the IV "shift toward home" is online (col. 4). Wind-IV caveats apply.

With the IV: a shift online, not a shorter shopping trip

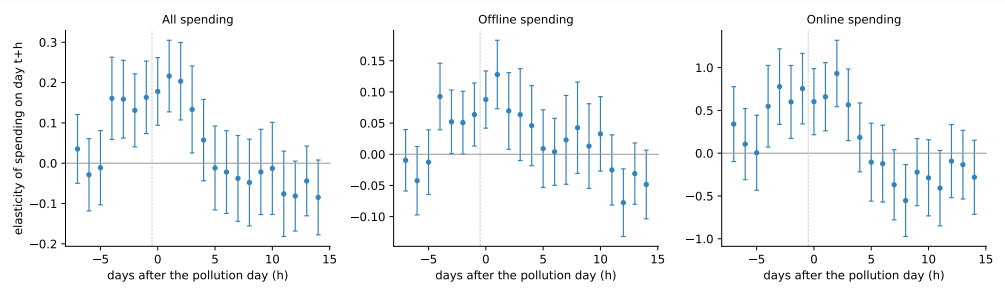
$S_{rt} = \beta \log \text{PM2.5}_{rt} + \alpha_r + X'_{rt}\gamma + \varepsilon_{rt}$ (OLS; 2SLS with wind sectors; weighted by mean spending)

Outcome	Mean	OLS	2SLS	KP F	AR p	AR 95% set
Home share (own dong + online)	0.223	-0.001	0.014**	11.3	0.030	[0.001, 0.033]
		(0.002)	(0.006)			
Ring 0 share	0.118	0.001	-0.009**	11.3	0.003	[-0.015, -0.005]
		(0.001)	(0.003)			
Ring 1 share	0.158	0.001	-0.006**	11.3	0.000	empty
		(0.001)	(0.003)			
Ring 2 share	0.108	0.000	-0.003**	11.3	0.501	[-0.006, 0.001]
		(0.000)	(0.001)			
Ring 3+ share	0.511	-0.000	-0.005**	11.3	0.624	[-0.015, 0.005]
		(0.001)	(0.003)			
Online share	0.105	-0.002	0.023***	11.3	0.005	[0.008, 0.048]
		(0.003)	(0.009)			
Mean distance, offline (km)	4.526	-0.014	0.064**	11.3	0.001	[0.078, 0.111]
		(0.011)	(0.032)			
Mean distance, online = 0 (km)	4.050	-0.002	-0.049*	11.3	0.754	[-0.153, 0.062]
		(0.011)	(0.029)			
Mean borders crossed, offline	3.978	-0.008	0.028	11.3	0.001	empty
		(0.006)	(0.018)			
HHI across merchant dongs	0.119	0.001	-0.001	11.3	0.001	empty
		(0.000)	(0.001)			
Merchant dongs visited	162.280	-0.125	2.381**	11.3	0.046	[0.066, 4.672]
		(0.327)	(0.955)			

Online share +2.3 pp per log point of PM2.5 (mean 10.5%); own-dong share falls; offline distance and number of places visited rise slightly.

Level effects fail the placebo: leads look like the effect

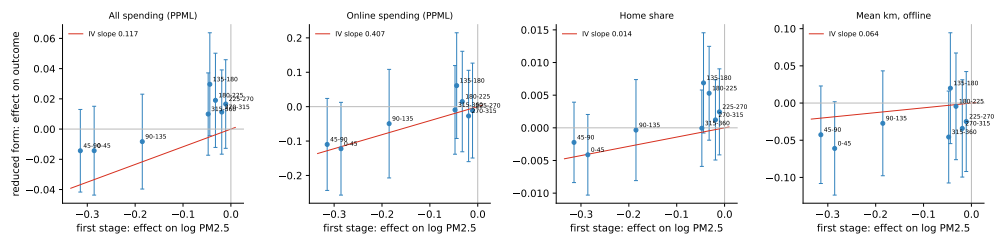
$$E[Y_{r,t+h}] = \exp\left(\beta_h \log \text{PM2.5}_{rt} + \rho_1 \log \text{PM2.5}_{r,t-1} + \rho_2 \log \text{PM2.5}_{r,t-2} + \alpha_r + X'_{rt}\gamma_h + \kappa_h \hat{v}_{rt}\right)$$



Wind-instrumented PM2.5 on day t "predicts" spending on days $t-4 \ldots t-1$ as much as on day t: persistent wind regimes. Do not use IV level effects.

Exclusion check: reduced form vs. first stage by sector

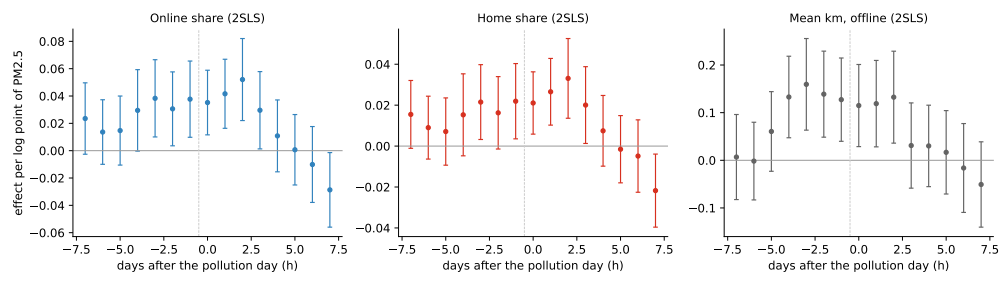
$$Y_{rt} = \sum_s \theta_s \mathbf{1}[\text{WD}_{rt} = s] + \alpha_r + X_{rt}'\gamma + u_{rt}; \quad \text{exclusion} \Rightarrow \theta_s = \beta\, \pi_s$$



Under exclusion, points lie on the red line. South-easterly winds barely move PM2.5 but move spending; Hansen J rejects for some outcomes.

The relative outcomes show the same lead pattern

$$S_{r,t+h} = \beta_h \log \mathrm{PM2.5}_{rt} + \rho_1 \log \mathrm{PM2.5}_{r,t-1} + \rho_2 \log \mathrm{PM2.5}_{r,t-2} + \alpha_r + X'_{rt}\gamma_h + \varepsilon_{r,t+h} \quad \text{(2SLS, wind sectors)}$$



Online share is ≈3 pp higher on the four days *before* the instrumented shock, as on day t; unchanged with wind on $t \pm 2$ controlled. The instrument dates multi-day episodes / weather regimes, not days ⇒ the online shift is suggestive.

Robustness of the main estimates

Rings 1, 3+: β_j in $E[Y_{rkt}] = \exp(\sum_j \beta_j \mathbf{1}[k=j] \log \text{PM2.5}_{rt} + \mu_{rt} + \psi_{rk} + X'_{rt}\lambda_k)$ (PPML-IV); home share, km: β in $S_{rt} = \beta \log \text{PM2.5}_{rt} + \alpha_r + X'_{rt}\gamma + \varepsilon_{rt}$ (2SLS)

	Ring 1 (stacked PPML-IV)	Ring 3+ (stacked PPML-IV)	Home share (2SLS)	Mean km (2SLS)
Main specification	-0.124*** (0.047)	-0.092** (0.038)	0.014** (0.006)	0.064** (0.032)
Exposure				
City-wide mean PM2.5	-0.125*** (0.047)	-0.093** (0.038)	0.015*** (0.006)	0.071** (0.033)
All reporting monitors (previous rule)	-0.123*** (0.046)	-0.092** (0.037)	0.014** (0.006)	0.064** (0.032)
Nearest monitor only	-0.124*** (0.046)	-0.093** (0.037)	0.014** (0.005)	0.064** (0.031)
10 km radius	-0.123*** (0.047)	-0.092** (0.038)	0.014** (0.006)	0.066** (0.032)
Seoul monitors only (25)	-0.123*** (0.046)	-0.092** (0.037)	0.014** (0.006)	0.064** (0.032)
Roadside monitors included	-0.126*** (0.047)	-0.094** (0.038)	0.015** (0.006)	0.067** (0.032)
AQI instead of PM2.5	-0.243*** (0.081)	-0.170** (0.066)	0.028*** (0.010)	0.138** (0.058)
Controls				
No weather controls	-0.066** (0.031)	-0.043* (0.025)	0.010** (0.004)	0.062** (0.024)
Max × min temperature cells	-0.075 (0.045)	-0.046 (0.037)	0.009 (0.006)	0.057* (0.031)
Leads and lags of weather	-0.120** (0.050)	-0.090** (0.040)	0.013** (0.006)	0.063* (0.032)
Leads and lags of wind direction	-0.114* (0.059)	-0.086* (0.048)	0.012 (0.007)	0.046 (0.042)
Lagged PM2.5	-0.200*** (0.067)	-0.137** (0.055)	0.021*** (0.008)	0.115*** (0.044)
Instruments				
Wind sector × alert area (40)	-0.077** (0.033)	-0.058** (0.026)	0.014*** (0.005)	0.061** (0.030)
City-wide wind sector	-0.107** (0.047)	-0.076** (0.038)	0.015*** (0.006)	0.084*** (0.031)
LIML, wind × area			0.014*** (0.005)	0.062** (0.031)
Sample and weights				
Excluding the windiest decile of days	-0.153*** (0.055)	-0.116*** (0.045)	0.016** (0.006)	
2018 only	-0.069* (0.036)	-0.054* (0.029)	0.008 (0.005)	
2019 only	-0.346*** (0.109)	-0.261*** (0.098)	0.026* (0.014)	
Excluding holidays	-0.110** (0.047)	-0.079** (0.038)	0.012** (0.006)	
Weekdays	-0.123** (0.057)	-0.095** (0.045)	0.016** (0.008)	
Weekends	-0.012 (0.033)	0.020 (0.030)	-0.002 (0.003)	
Unweighted			0.018** (0.007)	0.096** (0.046)
Small dongs (below median area)	-0.317** (0.126)	-0.255** (0.124)	0.033** (0.013)	
Large dongs	-0.084** (0.039)	-0.059* (0.031)	0.013** (0.005)	

Inference: clustering by date is what matters

Standard errors of β in the level (all spending), stacked (ring 3+) and share/km equations above

Clustering	All spending	Ring 3+	Home share	Mean km
Residence + date (main)	0.0306 [0.000]	0.0376 [0.014]	0.0056 [0.011]	0.0317 [0.042]
Residence + year-week	0.0385 [0.003]	0.0402 [0.025]	0.0067 [0.037]	0.0359 [0.077]
District (25) + date	0.0341 [0.002]	0.0385 [0.024]	0.0055 [0.016]	0.0321 [0.056]
Date only	0.0282 [0.000]	0.0370 [0.013]	0.0056 [0.011]	0.0321 [0.045]
Heteroskedasticity-robust only	0.0101 [0.000]	0.0123 [0.000]	0.0018 [0.000]	0.0124 [0.000]
Driscoll-Kraay, 14 lags			0.0067 [0.035]	0.0381 [0.091]
District + date, wild bootstrap			$p = 0.003$	$p = 0.043$

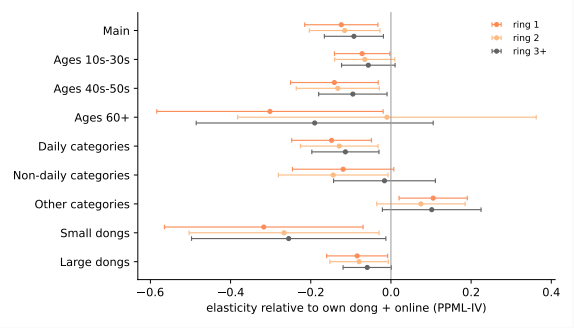
Robust-only SEs are about 3× too small. Serial correlation (year-week clusters, Driscoll–Kraay) adds 10–25%.

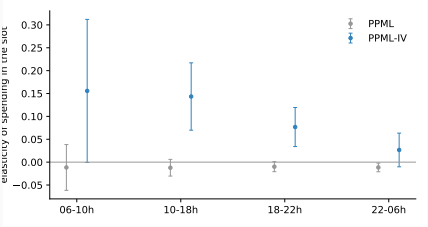
25 districts: $t(24)$ and wild bootstrap agree.

Who shifts

left: $E[Y^g_{rkt}] = \exp\left(\sum_{j\geq 1} \beta^g_j \mathbf{1}[k=j] \log \text{PM2.5}_{rt} + \mu^g_{rt} + \psi^g_{rk} + X'_{rt}\lambda^g_k\right)$ by group g

right: $E[Y^s_{rt}] = \exp(\beta^s \log \text{PM2.5}_{rt} + \alpha_r + X'_{rt}\gamma)$ by time slot s





Stronger in small (dense, central) dongs, ages 40–59, "daily" categories, weekdays; none on weekends.

Takeaways and next steps

Takeaways

- No contraction of the *offline* shopping map (city and panel, OLS and IV).
- Level effects are small: city OLS -0.8% per 10 $\mu g/m^3$, mostly online.
- Online response not identified: sign depends on the instrument; all fail lead placebos.
- Inference: cluster on time (Abadie et al.); NW, week, month clusters agree.

Next steps (to discuss)

- Episode-level event studies; air-quality alerts as information shocks.
- Stronger inversion measure (more soundings / reanalysis, inversion hours) with weather-matched placebos.
- District-level origin–destination flows; mobility data rebuilt from raw.

Backup: spending levels by ring

$$E[Y_{rt}] = \exp(\beta \log \text{PM2.5}_{rt} + \alpha_r + X'_{rt}\gamma) \quad \text{(PPML; PPML-IV adds } \hat{v}_{rt})$$

	All	Offline	Online	Home	Ring 0	Ring 1	Ring 2	Ring 3+
PPML: log PM2.5	-0.011	-0.004	-0.059*	-0.023	0.000	-0.002	-0.001	-0.008
	(0.008)	(0.005)	(0.033)	(0.014)	(0.005)	(0.006)	(0.005)	(0.005)
PPML-IV: log PM2.5	0.117***	0.057***	0.407***	0.200***	0.034**	0.046**	0.067***	0.074***
	(0.031)	(0.017)	(0.141)	(0.060)	(0.017)	(0.019)	(0.020)	(0.019)
First-stage F				6.9				

Backup: alternative instruments

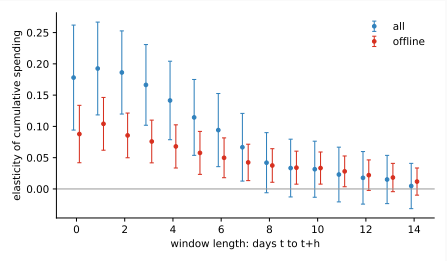
Same equations as the headline, levels and shares; only the instruments Z_{rt} change

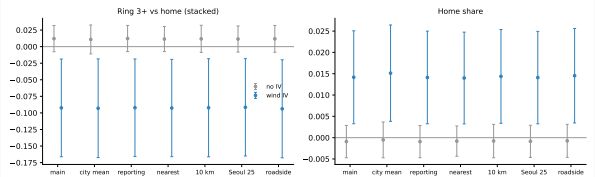
	All	Offline	Online	Ring 3+ vs home	Home share	Mean km	FS F
Wind sectors (8), main	0.117***	0.057***	0.407***	-0.092**	0.014**	0.064**	6.9
	(0.031)	(0.017)	(0.141)	(0.038)	(0.006)	(0.032)	
Westerly wind dummy (just identified)	0.098**	0.062***	0.255	-0.064	0.010	0.020	34.0
	(0.038)	(0.022)	(0.167)	(0.048)	(0.007)	(0.036)	
Wind sector × season (24)	0.057**	0.025*	0.215*	-0.025	0.008*	0.068**	4.0
	(0.024)	(0.014)	(0.111)	(0.028)	(0.004)	(0.028)	
Wind sector × alert area (40)				-0.058**	0.014***	0.061**	5.0
				(0.026)	(0.005)	(0.030)	
City-wide wind sector (8)				-0.076**	0.015***	0.084***	6.7
				(0.038)	(0.006)	(0.031)	

Backup: cumulative local projections and exposure measures

left: $E\left[\sum_{j=0}^{h} Y_{r,t+j}\right] = \exp(\beta_h \log \text{PM2.5}_{rt} + \rho_1 \log \text{PM2.5}_{r,t-1} + \rho_2 \log \text{PM2.5}_{r,t-2} + \alpha_r + X'_{rt}\gamma_h + \kappa_h \hat{v}_{rt})$

right: stacked ring-3+ and home-share estimates with PM2.5_{rt} built by alternative monitor rules





Exposure rule barely matters (96% of variation is city-wide). Full specification list:

`output/spatial/spec_batch1.pdf`.